

Shell Scripting Project: Log Rotation, Backup & Crontab

❖ Create log_rotate.sh that:

- Takes a log directory as an argument (eg./var/log/myapp)
- Compresses .log files older than 7 days using gzip
- Deletes .gz file files older than 30 days
- Prints how many files were compressed and deleted
- Exits with with an error if the directory doesn't exist
 - `#!/bin/bash`
 - `set -euo pipefail`
 - `LOG_DIR="${1:-}"`
 - `if [ -z "$LOG_DIR" ]; then`
 - `echo "Usage: $0 <log_directory>"`
 - `exit 1`
 - `fi`
 - `if [ ! -d "$LOG_DIR" ]; then`
 - `echo "Error: Directory does not exist"`
 - `exit 1`
 - `fi`
 - `echo "Processing logs in $LOG_DIR"`
 - `# Compress .log files older than 7 days`
 - `COMPRESSED_COUNT=$(find "$LOG_DIR" -type f -name "*.log" -mtime +7 -print -exec gzip {} \; | wc -l)`
 - `# Delete .gz files older than 30 days`
 - `DELETED_COUNT=$(find "$LOG_DIR" -type f -name "*.gz" -mtime +30 -print -delete | wc -l)`
 - `echo "Compressed files: $COMPRESSED_COUNT"`
 - `echo "Deleted old archives: $DELETED_COUNT"`

```
ubuntu@ip-172-31-47-149:~$ mkdir /tmp/test_logs
ubuntu@ip-172-31-47-149:~$ touch /tmp/test_logs/app.log
ubuntu@ip-172-31-47-149:~$ touch -d "10 days ago" /tmp/test_logs/app.log
ubuntu@ip-172-31-47-149:~$ ./log_rotate.sh /tmp/test_logs
Processing logs in /tmp/test_logs
Compressed files: 1
Deleted old archives: 0
```

❖ Task 2: Server Backup Script

- Create backup.sh that:
- Takes a source directory and backup destination as arguments
- Creates a timestamped .tar.gz archive (eg backup-2026-02-08.tar.gz)
- Verifies the archive was created successfully
- Prints archive name and size
- Deletes backups older than 14 days from the destination
- Handles errors – exit if source doesn't exist
 - `#!/bin/bash`
 - `set -euo pipefail`
 - `SOURCE="${1:-}"`
 - `DEST="${2:-}"`
 -
 - `if [ -z "$SOURCE" ] || [ -z "$DEST" ]; then`
 - `echo "Usage: $0 <source_directory> <backup_destination>"`
 - `exit 1`
 - `fi`
 -
 - `if [ ! -d "$SOURCE" ]; then`
 - `echo "Error: Source directory does not exist"`
 - `exit 1`
 - `fi`
 -
 - `mkdir -p "$DEST"`
 -
 - `TIMESTAMP=$(date +%Y-%m-%d)`
 - `ARCHIVE_NAME="backup-$TIMESTAMP.tar.gz"`
 - `ARCHIVE_PATH="$DEST/$ARCHIVE_NAME"`
 -
 - `echo "Creating backup..."`
 -
 - `tar -czf "$ARCHIVE_PATH" "$SOURCE"`
 -
 - `if [ -f "$ARCHIVE_PATH" ]; then`
 - `echo "Backup created successfully"`

- echo "Archive: \$ARCHIVE_NAME"
- du -h "\$ARCHIVE_PATH"
- else
- echo "Backup failed"
- exit 1
- fi
-
- # Delete backups older than 14 days
- find "\$DEST" -type f -name "backup-*.tar.gz" -mtime +14 -delete

```
ubuntu@ip-172-31-47-149:~$ ./backup.sh logs backup
Creating backup...
Backup created successfully
Archive: backup-2026-02-22.tar.gz
4.0K    backup/backup-2026-02-22.tar.gz
ubuntu@ip-172-31-47-149:~$
```

❖ Task 3 – Crontab

- Read crontab -l – what is currently scheduled?
- Under cron syntax:



- Write cron entries (in your markdown, don't apply if unsure) for:
 - Run log_rotate.sh day at 2 AM
 - 0 2 * * * /home/ubuntu/log_rotate.sh
/tmp/testlogs/myapp

- Run backup.sh every Sunday 3 AM
 - 0 3 * * 0 /home/ubuntu/backup.sh /home/ubuntu/logs /backup
- Run a health Check script every 5 minutes
 - */5 * * * * /home/ubuntu/health_check.sh

```
0 2 * * * /home/ubuntu/log_rotate.sh /tmp/testlogs/myapp
0 3 * * 0 /home/ubuntu/backup.sh /home/ubuntu/logs /backup
*/5 * * * * /home/ubuntu/health_check.sh
```

- Task 4: Combined Maintenance Script
- Create maintenance.sh that:
 - Calls your log rotation function
 - Calls your backup function
 - Logs all output to /var/log/maintenance.log with timestamps
 - Write the cron entry to run it daily at 1 AM
 - #!/bin/bash
 - set -euo pipefail
 -
 - LOG_FILE="/var/log/maintenance.log"
 - log() {
 - echo "\$(date '+%Y-%m-%d %H:%M:%S') - \$1" >>
 - "\$LOG_FILE"
 - }
 -
 - run_log_rotation() {
 - log "Starting log rotation"
 - /home/ubuntu/log_rotate.sh /home/ubuntu/logs/ >>
 - "\$LOG_FILE" 2>&1
 - log "Log rotation completed"
 - }
 - run_backup() {

- log "Starting backup"
- /home/ubuntu/backup.sh /home/ubuntu/logs
/home/ubuntu/backup >> "\$LOG_FILE" 2>&1
- log "Backup completed"
- }
- main() {
- log "Maintenance job started"
- run_log_rotation
- run_backup
- log "Maintenance job finished"
- }
- main

```
ubuntu@ip-172-31-47-149:~$ sudo ./maintenance.sh
ubuntu@ip-172-31-47-149:~$ cat /var/log/maintenance.log
2026-02-22 15:12:59 - Maintenance job started
2026-02-22 15:12:59 - Starting log rotation
Processing logs in /home/ubuntu/logs/
Compressed files: 0
```

- What I Learned
- Automating log cleanup prevents disk space issues
- Backups must be timestamped and auto cleaned
- Cron enables fully automated server maintenance